

Control Charts for Attributes

The p, np, c and u charts — for fraction nonconforming & nonconformities (defects)

Slides covered: **7-14, 27-29, 31-37, 40-47, 52-67**

Summary + all formulas & examples

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1 Attributes vs. Variables — the big picture

Attributes data classify each unit as conforming / nonconforming, or count defects. Two families of attribute charts:

p, np

Fraction / number **nonconforming**
(Binomial)

c, u

Number of **nonconformities** (defects)
(Poisson)

Key distinction

Nonconforming unit = a defective item (fails to meet specs).

Nonconformity = a single defect; one unit may contain several nonconformities.

⚠ Recurring exam rule

For *all four* attribute charts: if the computed **LCL < 0**, set **LCL = 0** (the chart then has only an upper control limit).

The p Chart — Fraction Nonconforming

The sample fraction nonconforming $\hat{p}$ follows the **binomial** distribution. Its mean and standard deviation are:

$$\mu_{\hat{p}} = p \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Case A — Standard Given (p known)

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Eq 7.6

$$UCL = p + 3\sqrt{\frac{p(1-p)}{n}}$$

$$CL = p$$

$$LCL = p - 3\sqrt{\frac{p(1-p)}{n}}$$

Case B — No Standard Given (estimate p)

slides 9-10

Take m preliminary samples (rule of thumb $m \geq 20$ or 25), each of size n .
With D_i nonconforming units in sample i :

$$\hat{p}_i = \frac{D_i}{n} \quad \bar{p} = \frac{\sum D_i}{m \cdot n} = \frac{\sum \hat{p}_i}{m}$$

Eq 7.8

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$CL = \bar{p}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$



Example 7.1 — Orange juice cans

$m = 30$ samples, $n = 50$ each. Total nonconforming $\Sigma D = 347 \Rightarrow \bar{p} = 347 / (30 \cdot 50) = 0.2313$.

Trial limits: $UCL = 0.2313 + 3(0.0596) = 0.4102$, $CL = 0.2313$, $LCL = 0.0524$.

Samples **15** (new raw material) and **23** (new, inexperienced operator) plotted above UCL $\rightarrow$ **assignable causes**. Removing them and recomputing on 28 samples ($\Sigma D = 301$):

Revised: $\bar{p} = 301 / (28 \cdot 50) = 0.2150 \rightarrow UCL = 0.3893$, $CL = 0.2150$, $LCL = 0.0407$.

The np Chart — Number Nonconforming

Instead of the *fraction*, plot the **number** nonconforming. Many non-statistical staff find it easier to read. (If p is unknown, use $\bar{p}$.)

Eq 7.13

$$UCL = np + 3\sqrt{np(1 - p)}$$

$$CL = np$$

$$LCL = np - 3\sqrt{np(1 - p)}$$



Example 7.2 — same OJ data

$\bar{p} = 0.2313$, $n = 50$: $CL = n\bar{p} = 50(0.2313) = \mathbf{11.565}$, $UCL = \mathbf{20.510}$, $LCL = \mathbf{2.620}$. In practice plot whole numbers (e.g. use 2 and 21 as LCL/UCL).

When sample size n changes from sample to sample, there are three approaches:

I. Variable-Width Control Limits

Compute limits separately for each sample using its own size n_i :

$$\bar{p} \pm 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

II. Control Limits Based on an Average Sample Size

Use one set of approximate limits from the average size (assumes future sizes won't differ greatly):

$$\bar{n} = \frac{\sum n_i}{m} \rightarrow \bar{p} \pm 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{\bar{n}}}$$

III. Standardized Control Chart

Plot points in standard-deviation units: center line = 0, UCL = +3, LCL = -3.

$$Z_i = \frac{\hat{p}_i - \bar{p}}{\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}}$$

The **operating-characteristic (OC) curve** plots the probability of a **type II (β) error** — incorrectly accepting "in control" — against the true fraction nonconforming. It measures the chart's *sensitivity* to a shift.

Eq 7.15 (β -error)

$$\begin{aligned}\beta &= P[\hat{p} < \text{UCL} \mid p] - P[\hat{p} \leq \text{LCL} \mid p] \\ &= P[D < n \cdot \text{UCL} \mid p] - P[D \leq n \cdot \text{LCL} \mid p]\end{aligned}$$

D is binomial(n, p), so β comes from the cumulative binomial. **If LCL is negative, drop the second term.**

Average Run Length (ARL)

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$$\text{ARL} = \frac{1}{P(\text{point plots out of control})}$$

$$\text{In control: } \text{ARL}_0 = \frac{1}{\alpha} \quad \cdot \quad \text{Out of control: } \text{ARL}_1 = \frac{1}{1 - \beta}$$

ARL example (slide 37)

$n = 50$, $\text{UCL} = 0.3697$, $\text{LCL} = 0.0303$, center $\bar{p} = 0.20$. From the OC curve $\alpha = 1 - 0.9973 = 0.0027$, so **$\text{ARL}_0 = 1/0.0027 \approx 370$** (a false alarm about every 370 samples).

If p shifts to 0.30, $\beta = 0.8594$, so **$\text{ARL}_1 = 1/(1 - 0.8594) \approx 7$** — about 7 samples to detect the shift.

The c Chart — Nonconformities (Defects)

Charts the **total number of defects** in an inspection unit, modeled by the **Poisson** distribution. This requires: (1) a very large number of opportunities for defects, (2) a small, constant probability of a defect at any location, and (3) the **same inspection unit** every sample.

$$p(x) = \frac{e^{-c} c^x}{x!}, \quad x = 0, 1, 2, \dots \quad (\text{mean} = \text{variance} = c)$$

Case A — Standard Given (c known)

Eq 7.16

$$UCL = c + 3\sqrt{c} \quad \cdot \quad CL = c \quad \cdot \quad LCL = c - 3\sqrt{c}$$

Case B — No Standard Given (estimate $\bar{c}$)

Eq 7.17

$$UCL = \bar{c} + 3\sqrt{\bar{c}} \quad \cdot \quad CL = \bar{c} \quad \cdot \quad LCL = \bar{c} - 3\sqrt{\bar{c}}$$



Example 7.3 — Printed circuit boards

26 samples, inspection unit = 100 boards. Total defects = 516 $\Rightarrow \bar{c} = 516/26 = 19.85$.

Trial: $UCL = 19.85 + 3\sqrt{19.85} = 33.22$, $CL = 19.85$, $LCL = 6.48$.

Samples **6** (new inspector) and **20** (temperature-control problem) were out of control $\rightarrow$ removed. **Revised:** $\bar{c} = 472/24 = 19.67 \rightarrow UCL = 32.97$, $LCL = 6.36$.



Further analysis

Defect data are richer than fraction nonconforming: classify defects *by type* and use a **Pareto chart** to find the "vital few" (e.g. solder insufficiency & cold joints) feeding the **out-of-control-action plans (OCAPs)**.

When the sample contains **n inspection units**, chart the **average defects per unit** $u = x / n$ (x is Poisson). The u chart should **always be used when the sample size is variable**.

Eq 7.19

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} \quad \cdot \quad CL = \bar{u} \quad \cdot \quad LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$



Example 7.4 — Supply-chain shipping errors

20 weeks, n = 50 shipments/week. Σ defects = 74 $\Rightarrow \bar{u} = \Sigma u_i / 20 = 1.48 / 20 = 0.0740$.

$UCL = 0.0740 + 3\sqrt{(0.0740/50)} = 0.1894$, $CL = 0.0740$, $LCL = 0.0740 - \dots = -0.0414 \rightarrow$ **set LCL = 0**.

Variable sample size on a u chart

Two options: (1) limits from average size $\bar{n} = \Sigma n_i / m$; or (2) the **standardized chart** (preferred), with $CL = 0$, $UCL = +3$, $LCL = -3$:

$$Z_i = \frac{u_i - \bar{u}}{\sqrt{\frac{\bar{u}}{n_i}}}$$

For complex products, not all defects are equally important, so classify them by **severity** and assign **weights**:

Class	Severity	Weight
A	Very serious (unfit / unsafe)	100
B	Serious	50
C	Moderately serious	10
D	Minor	1

Demerits in inspection unit i , and demerits per unit for a sample of n units ($D = \sum d_i$):

Eq 7.21-7.22

$$d_i = 100c_{iA} + 50c_{iB} + 10c_{iC} + c_{iD} \quad \cdot \quad u_i = \frac{D}{n}$$

Eq 7.23-7.25

$$CL = \hat{u} = 100\bar{u}_A + 50\bar{u}_B + 10\bar{u}_C + \bar{u}_D$$

$$UCL / LCL = \hat{u} \pm 3\hat{\sigma}_u$$

$$\hat{\sigma}_u = \sqrt{\frac{(100^2\bar{u}_A + 50^2\bar{u}_B + 10^2\bar{u}_C + 1^2\bar{u}_D)}{n}}$$

Attributes vs. Variables charts

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Attributes (p, np, c, u)	Variables ($\bar{x}$ -R, $\bar{x}$ -s)
Several characteristics judged jointly (conform / not)	More information about process performance
Often easier & cheaper to collect	Gives advance warning before defects occur
Usually needs larger samples	Needs smaller samples

Which chart type?

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- **$\bar{x}$ and R (or $\bar{x}$ and s):** new process; tight/overlapping tolerances; diagnosing a chronically troubled process; when destructive testing or reduced sampling is wanted.
- **Attributes (p, c, u):** operators control assignable causes; complex assembly; process control needed but measurement data unavailable; management summaries.
- **Individuals (x) + moving range:** one measurement per unit (e.g. chemical processes), or data available very slowly.

Actions taken to improve a process

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Decisions depend on two questions — *is the process in control?* and *is it capable?*

	Capable	Not capable
In control	SPC	SPC · Experimental design · Investigate specs · Change process
Not in control	SPC	SPC · Experimental design · Investigate specs · Change process



Placement guideline

The **earlier in the process that control can be established, the better**. As knowledge grows, charts often shift toward attributes near the *end* of manufacturing, while $\bar{x}$ -R charts move *earlier* to the critical parameters.



Formula Cheat Sheet

Chart	Center line	Control limits (3 σ)	Distribution
p (standard)	p	$p \pm 3\sqrt{[p(1-p)/n]}$	Binomial
p (no standard)	$\bar{p}$	$\bar{p} \pm 3\sqrt{[\bar{p}(1-\bar{p})/n]}$	Binomial
np	np	$np \pm 3\sqrt{[np(1-p)]}$	Binomial
c	$\bar{c}$ (or c)	$\bar{c} \pm 3\sqrt{\bar{c}}$	Poisson
u	$\bar{u}$	$\bar{u} \pm 3\sqrt{(\bar{u}/n)}$	Poisson

Quantity	Formula
Sample fraction (sample i)	$\hat{p}_i = D_i / n$
Average fraction	$\bar{p} = \Sigma D_i / (m \cdot n) = \Sigma \hat{p}_i / m$
Average sample size	$\bar{n} = \Sigma n_i / m$
Standardized p (variable n)	$Z_i = (\hat{p}_i - \bar{p}) / \sqrt{[\bar{p}(1-\bar{p})/n_i]}$
Standardized u (variable n)	$Z_i = (u_i - \bar{u}) / \sqrt{(\bar{u}/n_i)}$
β error	$\beta = P[D < n \cdot \text{UCL} \mid p] - P[D \leq n \cdot \text{LCL} \mid p]$
ARL (in / out of control)	$\text{ARL}_0 = 1/\alpha \quad \cdot \quad \text{ARL}_1 = 1/(1-\beta)$

Remember

If any LCL comes out negative, **set LCL = 0**. Use **m $\geq$ 20-25** preliminary samples to estimate parameters (phase I / trial limits).



Quick Review

- **Nonconforming (defective) → Binomial → p & np charts.**
- **Nonconformities (defects) → Poisson → c & u charts.**
- **p** = fraction, **np** = number, **c** = defects per (constant) inspection unit, **u** = defects per unit (and for *variable* sample size).
- **Variable n :** variable-width limits · average-size limits · standardized chart (CL 0, ± 3).
- **OC curve** shows β ; **$ARL_0 = 1/\alpha$, $ARL_1 = 1/(1-\beta)$** ; larger $n \Rightarrow$ faster detection.
- **Demerit weights:** $A = 100$, $B = 50$, $C = 10$, $D = 1$.
- Always **set LCL = 0 if negative**; estimate parameters from **$\geq 20-25$** samples.